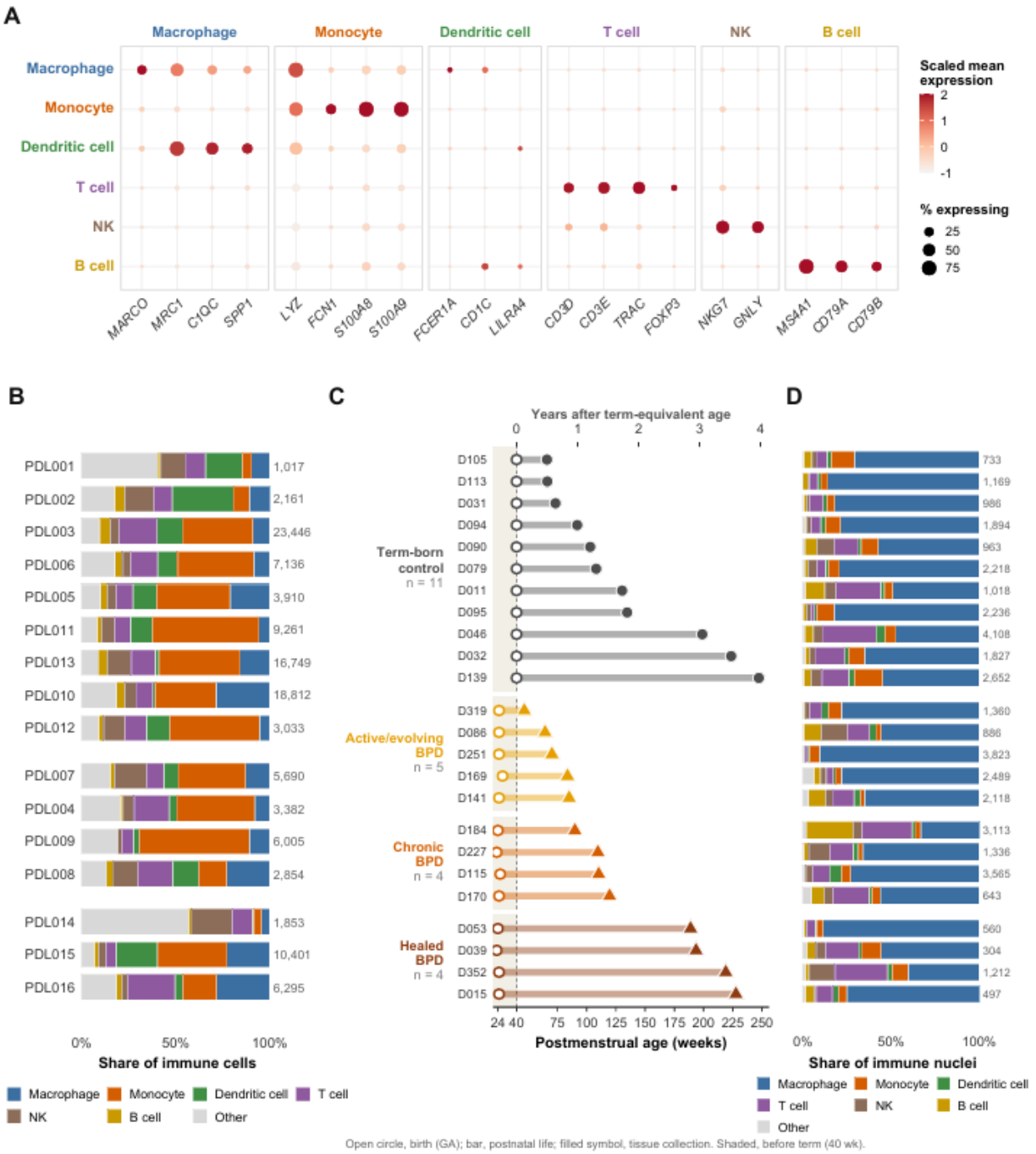
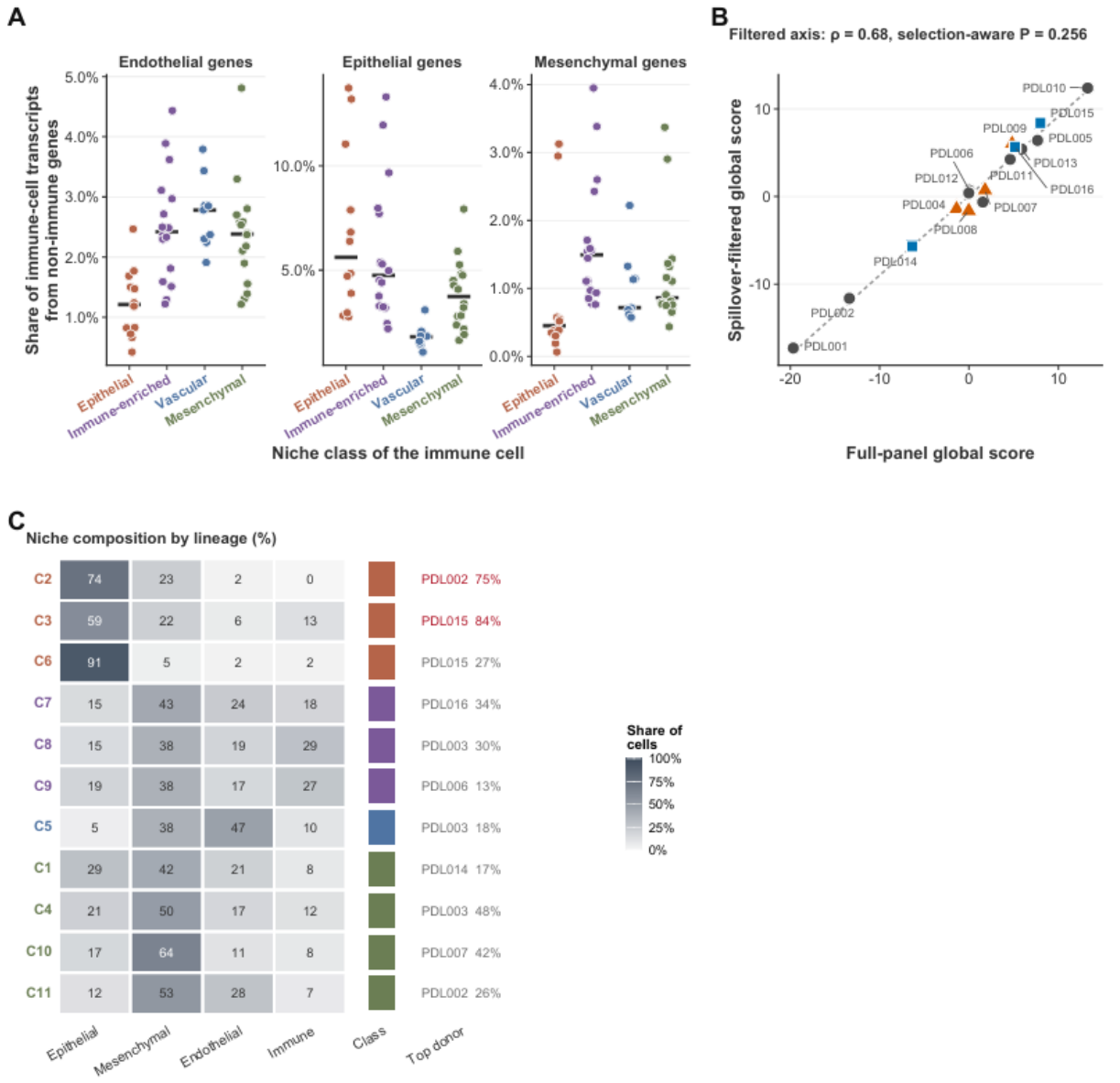
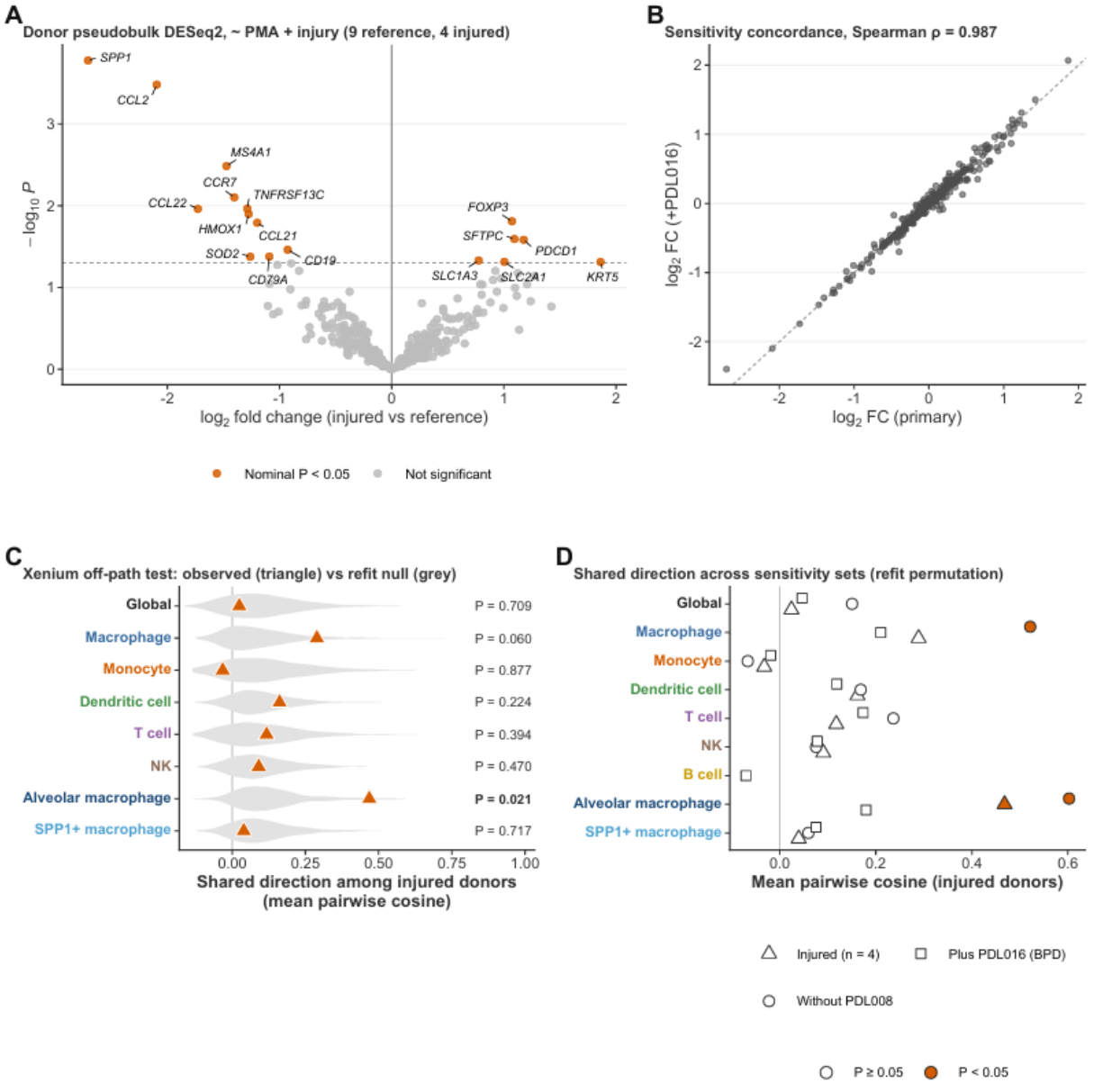
Supplemental Figure S1. Cohort details. A: Expression of canonical markers within the 343-gene panel across the six broad populations. Dot size, percentage of cells expressing; color, scaled mean expression. B: Xenium immune composition per donor; numbers give immune cells per donor, and Other denotes populations outside the six broad groups. C: LungMAP donor life courses against PMA: open circle, birth (gestational age); bar, postnatal life; filled symbol, tissue collection. Shading marks the period before term-equivalent age (40 weeks). D: LungMAP immune composition per donor; numbers give immune nuclei per donor.



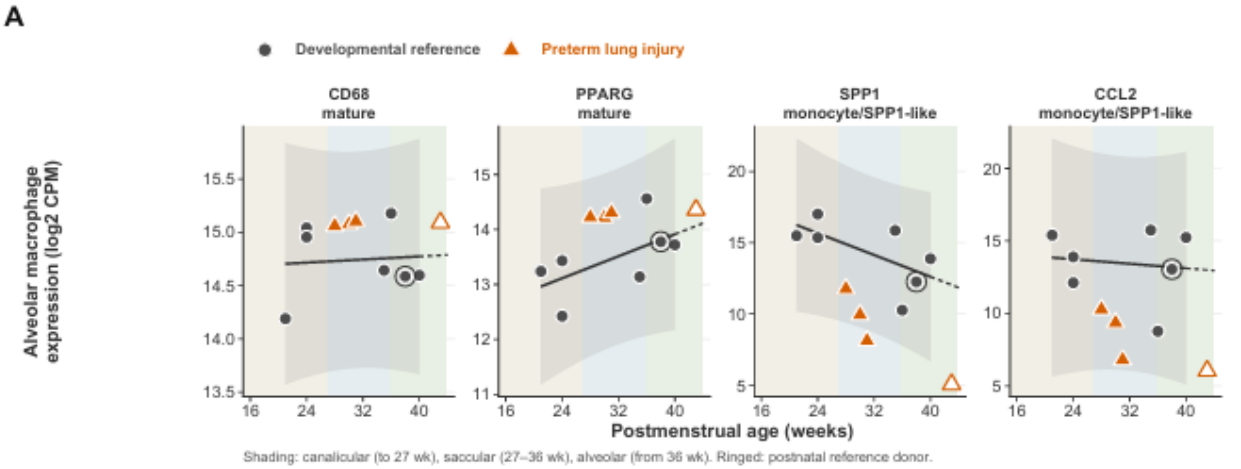
Supplemental Figure S2. Transcript spillover and niches. A: Share of each immune cell's transcripts derived from predominantly endothelial, epithelial or mesenchymal genes, by niche class of the immune cell. Points are donors; bars are medians. B: Spillover-filtered (77 non-immune genes removed) vs. full-panel global score; filtered axis ρ = 0.68, selection-aware P = 0.256. C: Lineage composition of the 11 source-defined niches, with the assigned niche class and the largest single-donor contribution.

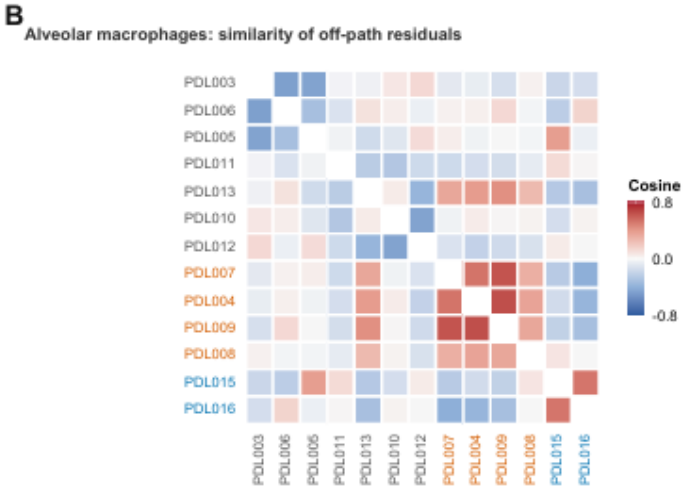


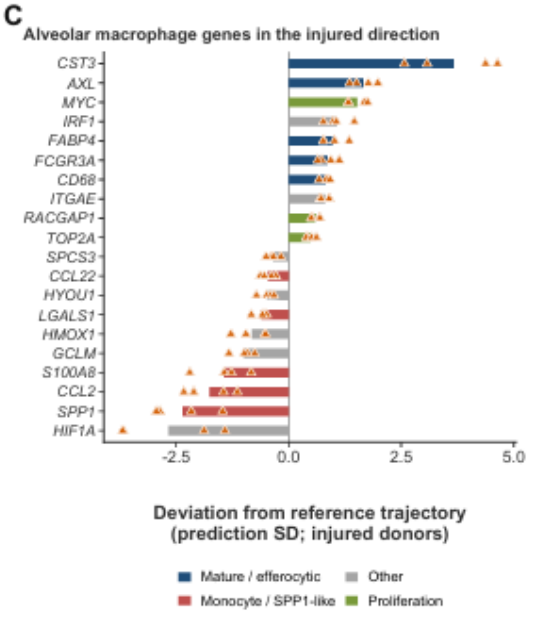
Supplemental Figure S3. Gene-level tests and shared off-path direction (Xenium). A: Donor pseudobulk DESeq2, ~ PMA + injury (9 reference, 4 injured). Labeled genes have nominal $P < 0.05$; none reached FDR < 0.05. B: Concordance of $\log_2$ fold changes with PDL016 added to the injury group (Spearman $\rho = 0.987$). C: Shared direction of departure among injured donors (mean pairwise cosine of standardized residuals); triangle, observed; grey, exact refit-permutation null. D: The same statistic across sensitivity sets (injured donors only; plus PDL016; without PDL008); filled symbols, $P < 0.05$.

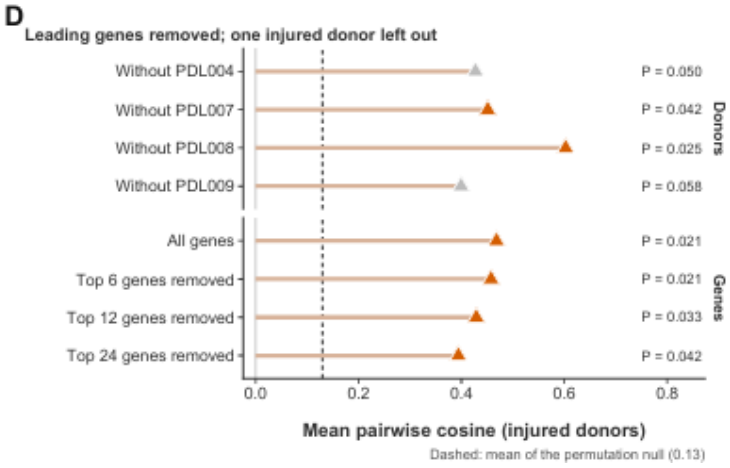


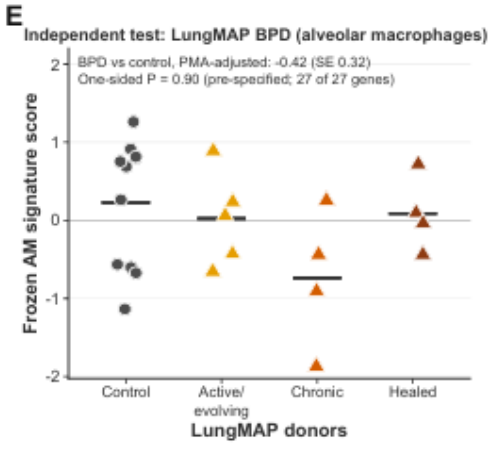
Supplemental Figure S4. Alveolar macrophages. A: Alveolar-macrophage expression of selected mature (*CD68, PPARG*) and monocyte/SPP1-like (*SPP1, CCL2*) genes against PMA. Line and band show the reference fit and 95% prediction interval; ringed, the postnatal reference donor. B: Cosine similarity of off-path residual vectors between donors. C: Genes contributing to the shared injured direction, as deviation from the reference trajectory in prediction-SD units, colored by gene category. D: Shared-direction statistic with one injured donor left out, and with the leading genes removed. The dashed line marks the mean of the permutation null. E: Frozen alveolar-macrophage signature tested in LungMAP alveolar macrophages by stage. Text gives the pre-specified one-sided test (BPD vs. control, PMA-adjusted).


A
Developmental reference
Preterm lung injury
CD68 mature
PPARG mature
SPP1 monocyte/SPP1-like
CCL2 monocyte/SPP1-like
Alveolar macrophage expression (log2 CPM)
Postmenstrual age (weeks)
Shading: canalicular (to 27 wk), saccular (27–36 wk), alveolar (from 36 wk). Ringed: postnatal reference donor.

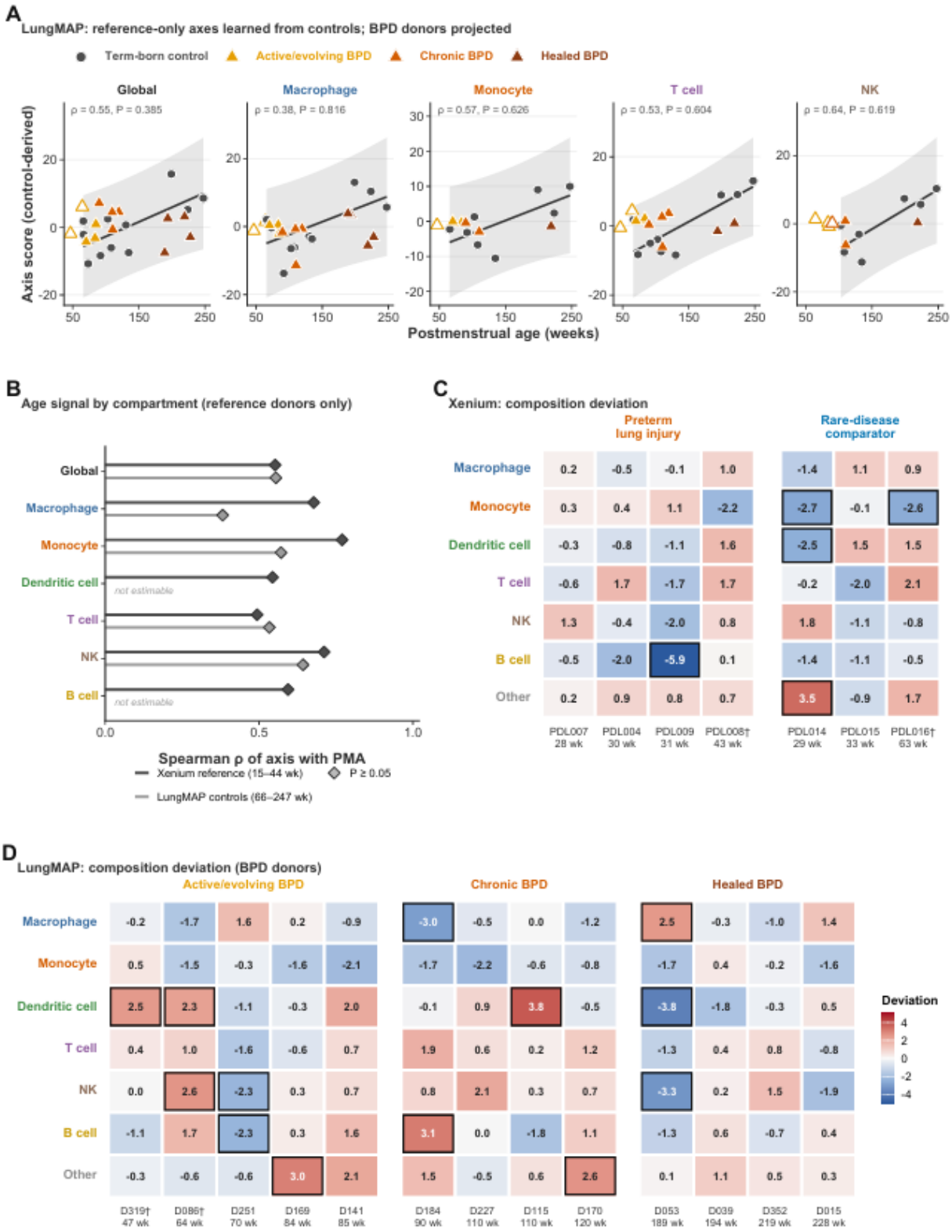

B
Alveolar macrophages: similarity of off-path residuals
PDL003
PDL006
PDL005
PDL011
PDL013
PDL010
PDL012
PDL007
PDL004
PDL009
PDL008
PDL015
PDL016
Cosine
0.8
0.0
-0.8


C
Alveolar macrophage genes in the injured direction
CST3
AXL
MYC
IRF1
FABP4
FCGR3A
CD68
ITGAE
RACGAP1
TOP2A
SPCS3
CCL22
HYOU1
LGALS1
HMOX1
GCLM
S100A8
CCL2
SPP1
HIF1A
Deviation from reference trajectory (prediction SD; injured donors)
Mature / efferocytic
Other
Monocyte / SPP1-like
Proliferation

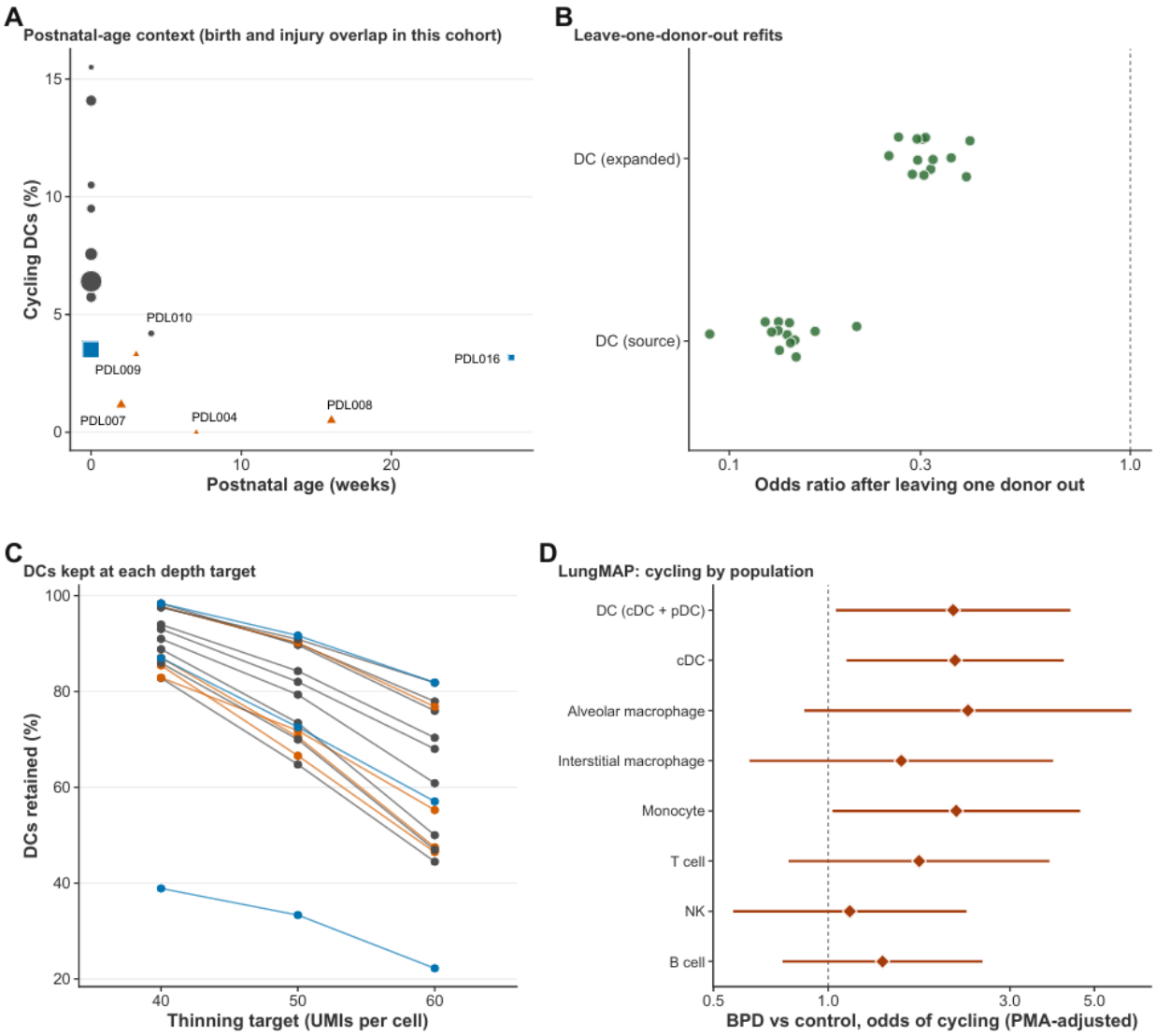

D
Leading genes removed; one injured donor left out
Without PDL004
Without PDL007
Without PDL008
Without PDL009
All genes
Top 6 genes removed
Top 12 genes removed
Top 24 genes removed
P = 0.050
P = 0.042
P = 0.025
P = 0.058
P = 0.021
P = 0.021
P = 0.033
P = 0.042
Donors
Genes
Mean pairwise cosine (injured donors)
Dashed: mean of the permutation null (0.13)


E
Independent test: LungMAP BPD (alveolar macrophages)
BPD vs control, PMA-adjusted: -0.42 (SE 0.32)
One-sided P = 0.90 (pre-specified; 27 of 27 genes)
Frozen AM signature score
Control
Active/ evolving
Chronic
Healed
LungMAP donors

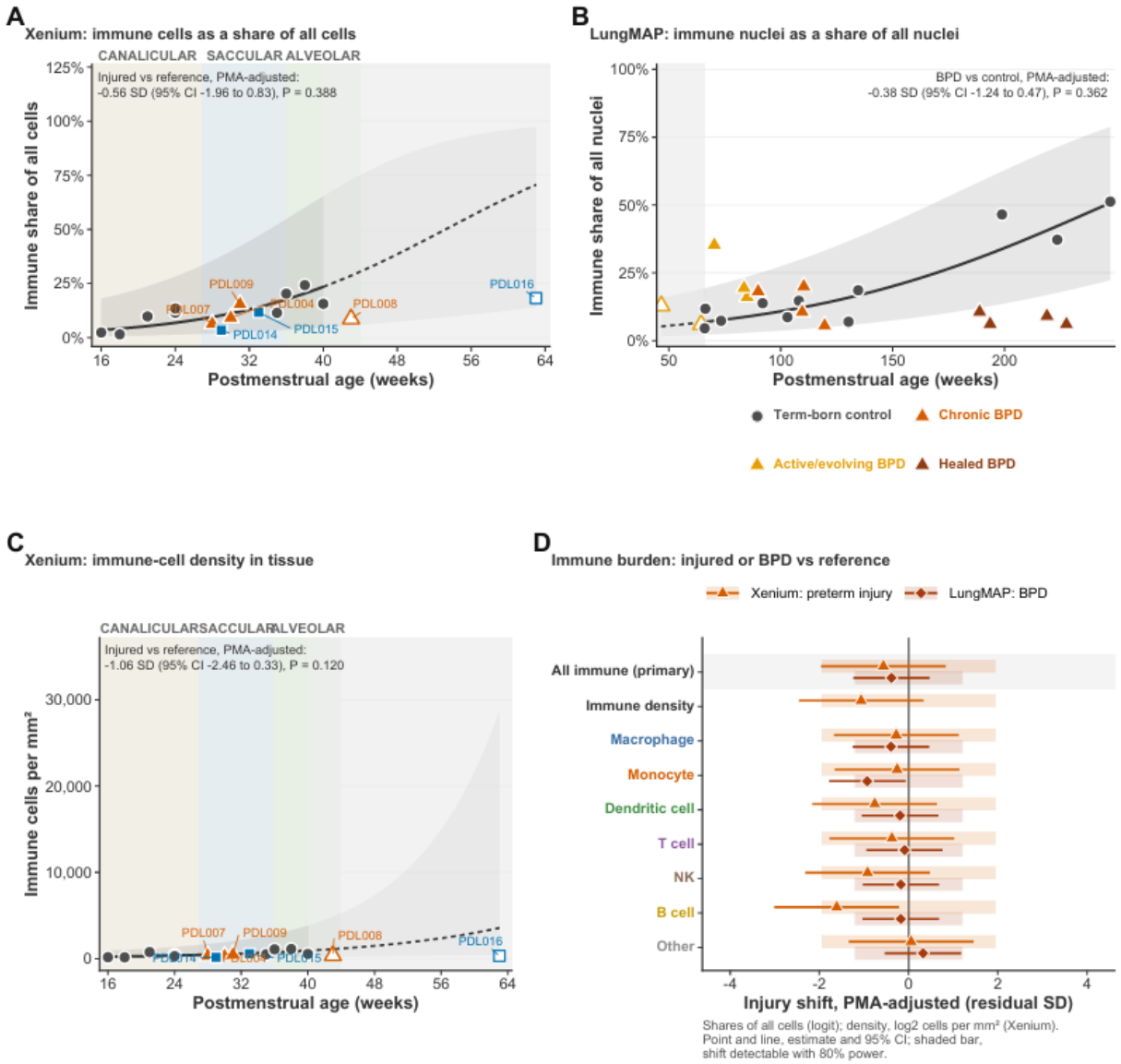
Supplemental Figure S5. LungMAP axes and composition deviations. A: LungMAP population axes learned from controls, against PMA, with BPD donors projected. Labels give ρ and selection-aware P. B: Strength of the age association by population: Xenium reference (15–44 weeks) vs. LungMAP controls (66–247 weeks); hollow symbols, $P \geq 0.05$. C: Xenium composition deviation of injured and rare-disease donors. D: LungMAP composition deviation of BPD donors by stage. In C and D, outlined tiles fall outside the 95% prediction limit; † marks extrapolation.


A
LungMAP: reference-only axes learned from controls; BPD donors projected
Term-born control
Active/evolving BPD
Chronic BPD
Healed BPD
Global
Macrophage
Monocyte
T cell
NK
ρ = 0.55, P = 0.385
ρ = 0.38, P = 0.816
ρ = 0.57, P = 0.626
ρ = 0.53, P = 0.604
ρ = 0.64, P = 0.619
Axis score (control-derived)
Postmenstrual age (weeks)
B
Age signal by compartment (reference donors only)
Global
Macrophage
Monocyte
Dendritic cell
T cell
NK
B cell
not estimable
Spearman ρ of axis with PMA
Xenium reference (15–44 wk)
P ≥ 0.05
LungMAP controls (66–247 wk)
C
Xenium: composition deviation
Preterm lung injury
Rare-disease comparator
Macrophage 0.2 -0.5 -0.1 1.0 -1.4 1.1 0.9
Monocyte 0.3 0.4 1.1 -2.2 -2.7 -0.1 -2.6
Dendritic cell -0.3 -0.8 -1.1 1.6 -2.5 1.5 1.5
T cell -0.6 1.7 -1.7 1.7 -0.2 -2.0 2.1
NK 1.3 -0.4 -2.0 0.8 1.8 -1.1 -0.8
B cell -0.5 -2.0 -5.9 0.1 -1.4 -1.1 -0.5
Other 0.2 0.9 0.8 0.7 3.5 -0.9 1.7
PDL007 28 wk
PDL004 30 wk
PDL009 31 wk
PDL008† 43 wk
PDL014 29 wk
PDL015 33 wk
PDL016† 63 wk
D
LungMAP: composition deviation (BPD donors)
Active/evolving BPD
Chronic BPD
Healed BPD
Macrophage -0.2 -1.7 1.6 0.2 -0.9 -3.0 -0.5 0.0 -1.2 2.5 -0.3 -1.0 1.4
Monocyte 0.5 -1.5 -0.3 -1.6 -2.1 -1.7 -2.2 -0.6 -0.8 -1.7 0.4 -0.2 -1.6
Dendritic cell 2.5 2.3 -1.1 -0.3 2.0 -0.1 0.9 3.8 -0.5 -3.8 -1.8 -0.3 0.5
T cell 0.4 1.0 -1.6 -0.6 0.7 1.9 0.6 0.2 1.2 -1.3 0.4 0.8 -0.8
NK 0.0 2.6 -2.3 0.3 0.7 0.8 2.1 0.3 0.7 -3.3 0.2 1.5 -1.9
B cell -1.1 1.7 -2.3 0.3 1.6 3.1 0.0 -1.8 1.1 -1.3 0.6 -0.7 0.4
Other -0.3 -0.6 -0.6 3.0 2.1 1.5 -0.5 0.6 2.6 0.1 1.1 0.5 0.3
D319† 47 wk
D086† 64 wk
D251 70 wk
D169 84 wk
D141 85 wk
D184 90 wk
D227 110 wk
D115 110 wk
D170 120 wk
D053 189 wk
D039 194 wk
D352 219 wk
D015 228 wk
Deviation
4
2
0
-2
-4

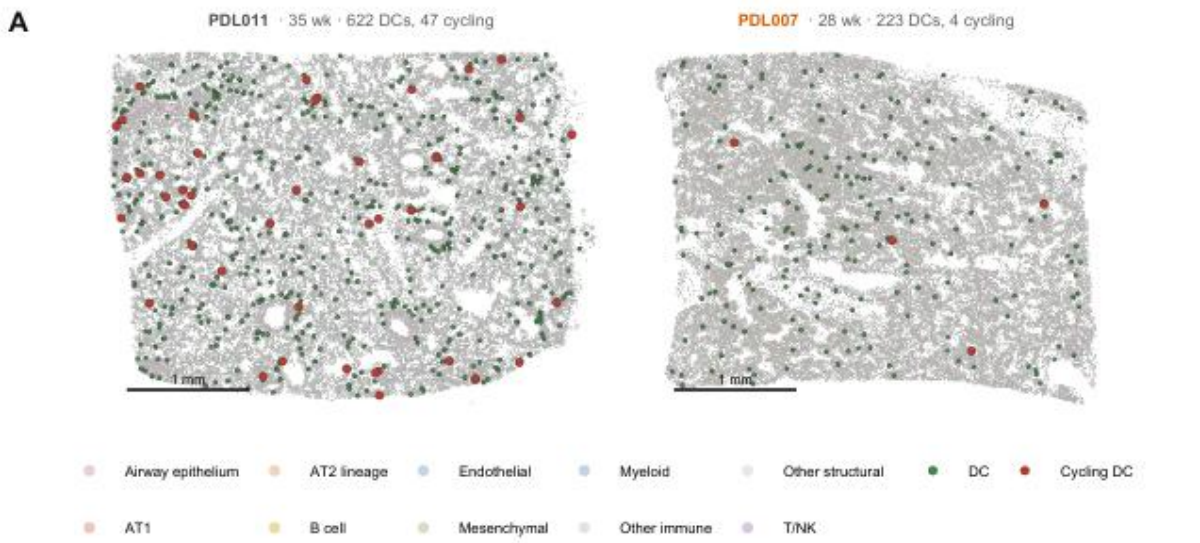
Supplemental Figure S6. Dendritic-cell cycling detail. A: Cycling DCs against postnatal age, showing that birth and injury overlap in Xenium (one postnatal reference donor). B: Odds ratios from leave-one-donor-out refits, for source-annotated and expanded DC definitions. C: Percentage of DCs retained at each rarefaction depth, per donor. D: LungMAP: odds of cycling by population, BPD vs. control (PMA-adjusted), with 95% confidence intervals.

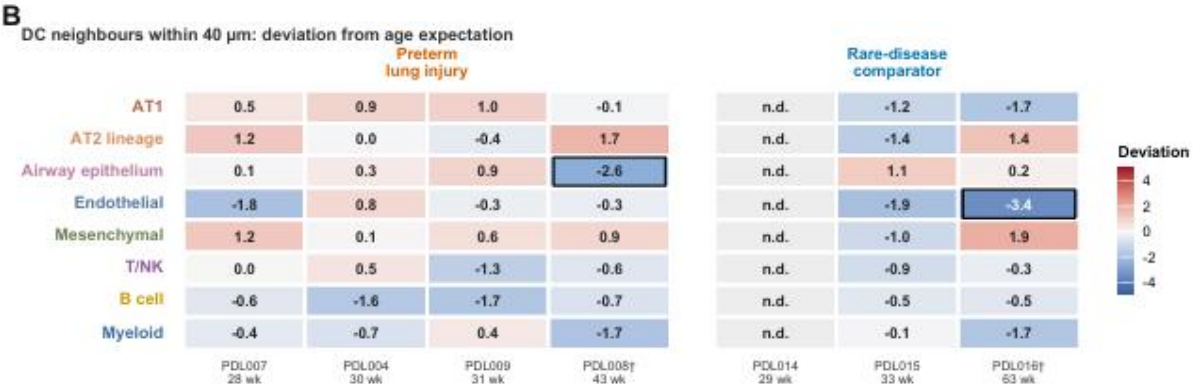


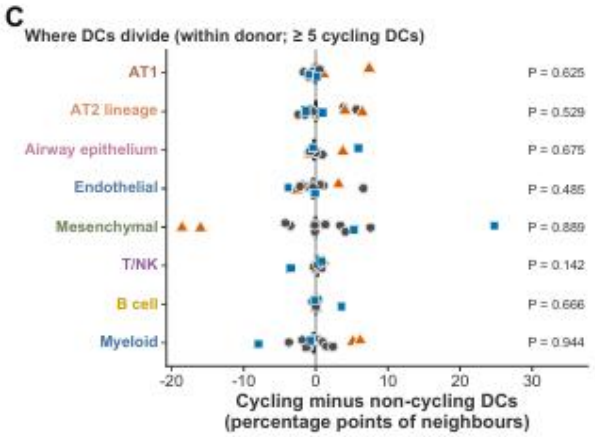
Supplemental Figure S7. Immune burden. A: Xenium: immune cells as a share of all cells, against PMA. B: LungMAP: immune nuclei as a share of all nuclei, against PMA. C: Xenium: immune-cell density per mm² of tissue. D: PMA-adjusted injury shifts for overall and population-specific burden, in residual-SD units. The shaded bar is the shift detectable with 80% power, and "All immune" is the primary measure. In A-C, the line and band show the reference or control fit and its 95% prediction interval, and text gives the PMA-adjusted shift.

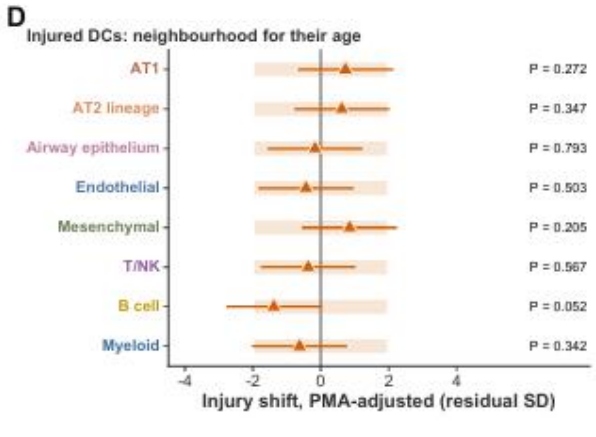


Supplemental Figure S8. Dendritic-cell localization (Xenium). A: Tissue maps of a reference donor and an injured donor; DCs (green) and cycling DCs (red) are overlaid on all other cells. Scale bar, 1 mm. B: Composition of cells within 40 μm of DCs, as deviation from age expectation; outlined tiles fall outside the 95% prediction limit, and † marks extrapolation. C: Within each donor with ≥5 cycling DCs, neighbor fraction of cycling minus non-cycling DCs. Bar, median; P, Wilcoxon signed-rank test. D: PMA-adjusted injury shift in DC neighbor composition; the shaded bar is the shift detectable with 80% power.

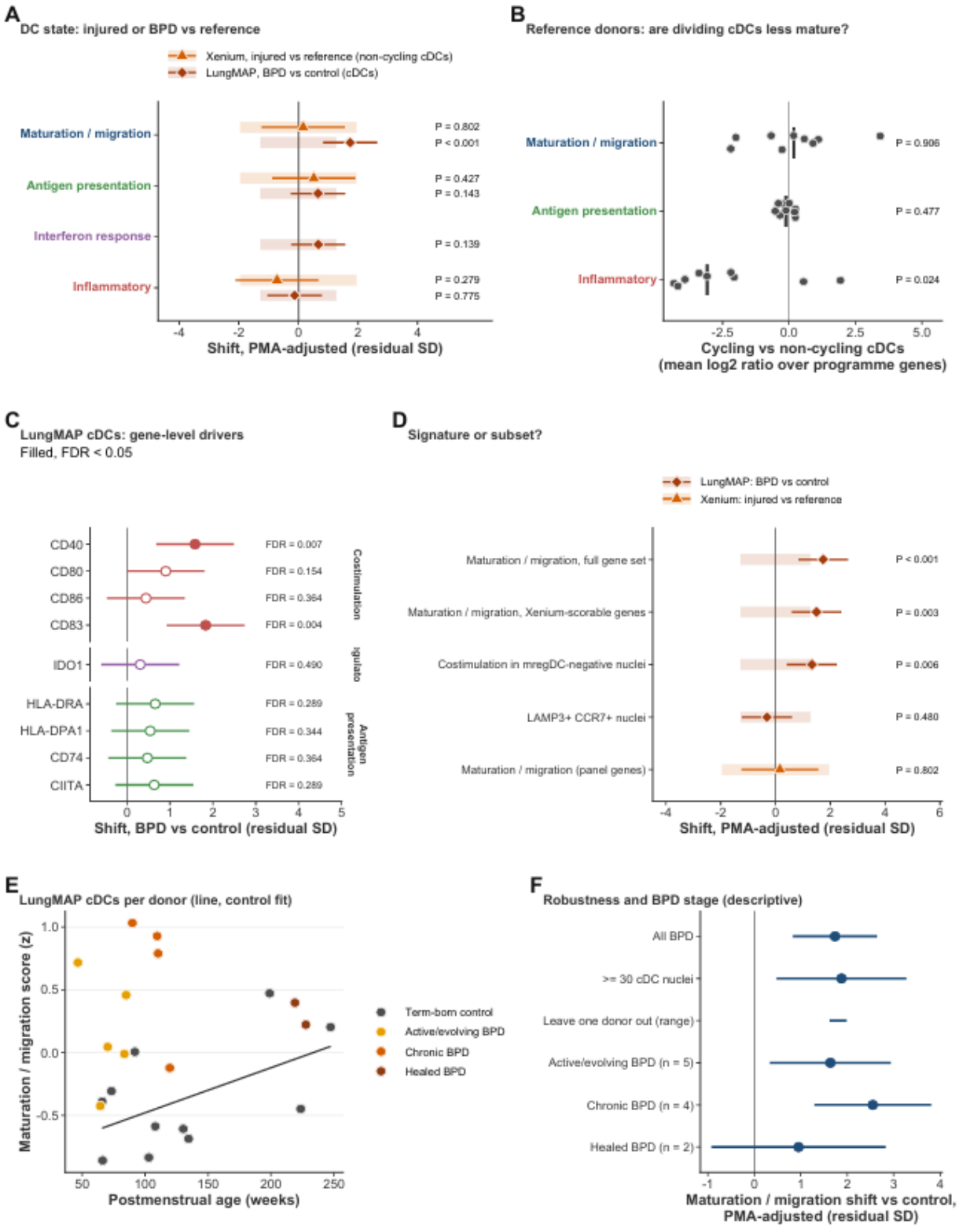

A
PDL011 · 35 wk · 622 DCs, 47 cycling
PDL007 · 28 wk · 223 DCs, 4 cycling
1 mm
1 mm
Airway epithelium
AT2 lineage
Endothelial
Myeloid
Other structural
DC
Cycling DC
AT1
B cell
Mesenchymal
Other immune
T/NK


B
DC neighbours within 40 µm: deviation from age expectation
Preterm lung injury
Rare-disease comparator
AT1 0.5 0.9 1.0 -0.1 n.d. -1.2 -1.7
AT2 lineage 1.2 0.0 -0.4 1.7 n.d. -1.4 1.4
Airway epithelium 0.1 0.3 0.9 -2.6 n.d. 1.1 0.2
Endothelial -1.8 0.8 -0.3 -0.3 n.d. -1.9 -3.4
Mesenchymal 1.2 0.1 0.6 0.9 n.d. -1.0 1.9
T/NK 0.0 0.5 -1.3 -0.6 n.d. -0.9 -0.3
B cell -0.6 -1.6 -1.7 -0.7 n.d. -0.5 -0.5
Myeloid -0.4 -0.7 0.4 -1.7 n.d. -0.1 -1.7
PDL007 28 wk
PDL004 30 wk
PDL009 31 wk
PDL008† 43 wk
PDL014 29 wk
PDL015 33 wk
PDL016† 63 wk
Deviation
4
2
0
-2
-4


C
Where DCs divide (within donor; ≥ 5 cycling DCs)
AT1 P = 0.625
AT2 lineage P = 0.529
Airway epithelium P = 0.675
Endothelial P = 0.485
Mesenchymal P = 0.889
T/NK P = 0.142
B cell P = 0.666
Myeloid P = 0.944
-20
-10
0
10
20
30
Cycling minus non-cycling DCs
(percentage points of neighbours)


D
Injured DCs: neighbourhood for their age
AT1 P = 0.272
AT2 lineage P = 0.347
Airway epithelium P = 0.793
Endothelial P = 0.503
Mesenchymal P = 0.205
T/NK P = 0.567
B cell P = 0.052
Myeloid P = 0.342
-4
-2
0
2
4
Injury shift, PMA-adjusted (residual SD)

Supplemental Figure S9. Dendritic-cell transcriptional state (exploratory; gene sets fixed before scoring). A: PMA-adjusted shift in four program scores, in residual-SD units: maturation/migration, antigen presentation, interferon response and inflammation (cell-cycle genes excluded; Xenium restricted to immune-specific panel genes). Shown for Xenium non-cycling conventional DCs (injured vs. reference) and LungMAP conventional DCs (BPD vs. control); shaded bar, shift detectable with 80% power. B: Xenium reference donors: mean $\log_2$ ratio between cycling and non-cycling conventional DCs within each donor. Bar, median; P, Wilcoxon signed-rank test. C: LungMAP conventional DCs: gene-level shifts for genes detected in ≥2/3 of donors; filled, FDR < 0.05. D: Maturation/migration score with the full gene set and with Xenium-scorable genes only; costimulation in nuclei without mature regulatory DC core genes; $LAMP3^{+}CCR7^{+}$ nuclei; and Xenium for comparison. E: LungMAP per-donor maturation/migration score against PMA; line, control fit. F: Robustness and stage (descriptive): all BPD donors, donors with ≥30 conventional-DC nuclei, the range of leave-one-donor-out estimates, and each stage vs. control.


A
DC state: injured or BPD vs reference
Xenium, injured vs reference (non-cycling cDCs)
LungMAP, BPD vs control (cDCs)
Maturation / migration
Antigen presentation
Interferon response
Inflammatory
P = 0.802
P < 0.001
P = 0.427
P = 0.143
P = 0.139
P = 0.279
P = 0.775
Shift, PMA-adjusted (residual SD)
B
Reference donors: are dividing cDCs less mature?
Maturation / migration
Antigen presentation
Inflammatory
P = 0.906
P = 0.477
P = 0.024
Cycling vs non-cycling cDCs
(mean log2 ratio over programme genes)
C
LungMAP cDCs: gene-level drivers
Filled, FDR < 0.05
CD40
CD80
CD86
CD83
IDO1
HLA-DRA
HLA-DPA1
CD74
CIITA
FDR = 0.007
FDR = 0.154
FDR = 0.364
FDR = 0.004
FDR = 0.490
FDR = 0.289
FDR = 0.344
FDR = 0.364
FDR = 0.289
Costimulation
Antigen presentation
Shift, BPD vs control (residual SD)
D
Signature or subset?
LungMAP: BPD vs control
Xenium: injured vs reference
Maturation / migration, full gene set
Maturation / migration, Xenium-scorable genes
Costimulation in mregDC-negative nuclei
LAMP3+ CCR7+ nuclei
Maturation / migration (panel genes)
P < 0.001
P = 0.003
P = 0.006
P = 0.480
P = 0.802
Shift, PMA-adjusted (residual SD)
E
LungMAP cDCs per donor (line, control fit)
Maturation / migration score (z)
Postmenstrual age (weeks)
Term-born control
Active/evolving BPD
Chronic BPD
Healed BPD
F
Robustness and BPD stage (descriptive)
All BPD
>= 30 cDC nuclei
Leave one donor out (range)
Active/evolving BPD (n = 5)
Chronic BPD (n = 4)
Healed BPD (n = 2)
Maturation / migration shift vs control,
PMA-adjusted (residual SD)